

deviate from an optimally prescribed schedule without adversely affecting performance. The recommendations are to accelerate trading up to the maximum trading opportunity in times of favorable market prices and liquidity and to decelerate trading to the minimum risk quantity in times of high impact costs and lower liquidity.

For example, whenever faced with favorable prices algorithmic trading rules can be specified to take advantage of the displayed liquidity up to the maximum trading opportunity. Algorithmic trading rules can also be specified to enter and display limit orders up to the maximum trading opportunity. In times of illiquidity, high market impact, or short-term price drift (with expected trend reversal) algorithmic trading rules can be written to decelerate trading down to the minimum trading risk quantity.

Program-Block Decomposition

When investors enter orders into crossing venues or dark pools the executions are not guaranteed. Transactions will occur only if there is a counterparty. When entering baskets into dark pools traders are often concerned that only some of their orders will trade and the resulting residual risk will be more than the original value.

One way investors can address this problem is to decompose the basket into block and program subsets. The block subset represents those shares contributing incremental risk to the basket. These are the shares that can be entered into a dark pool and would result in less risk no matter how many shares are executed. The program subset represents those shares that are providing risk reduction through either diversification or hedging. Accelerated trading of any of these shares is not recommended because if the executions are not in the proper proportions the resulting residual risk will be higher than the starting level of risk.

For example, stock A and stock B are perfectly correlated. If we are buying \$150K of A and selling \$100K of B we have \$50K worth of incremental risk from stock A. Thus, we could enter \$50K shares of A into a dark pool without worrying about our residual risk increasing. No matter how many shares of A trade in the dark pool the resulting portfolio risk will be lower than the original value. Now suppose that we are buying \$100K of stock A and selling \$100K of stock B (the same value in both stocks). Since these stocks are perfectly correlated our market exposure is hedged and the total portfolio risk is equal to the stock's idiosyncratic risk values. We want to trade these names together to minimize risk and maintain our hedged position. If we enter both stocks into a dark pool but are only able to transact one of the names then the resulting residual risk